ADVANCED DATABASE SYSTEMS PORTFOLIO3

MIT 8103 - Advanced Database Systems

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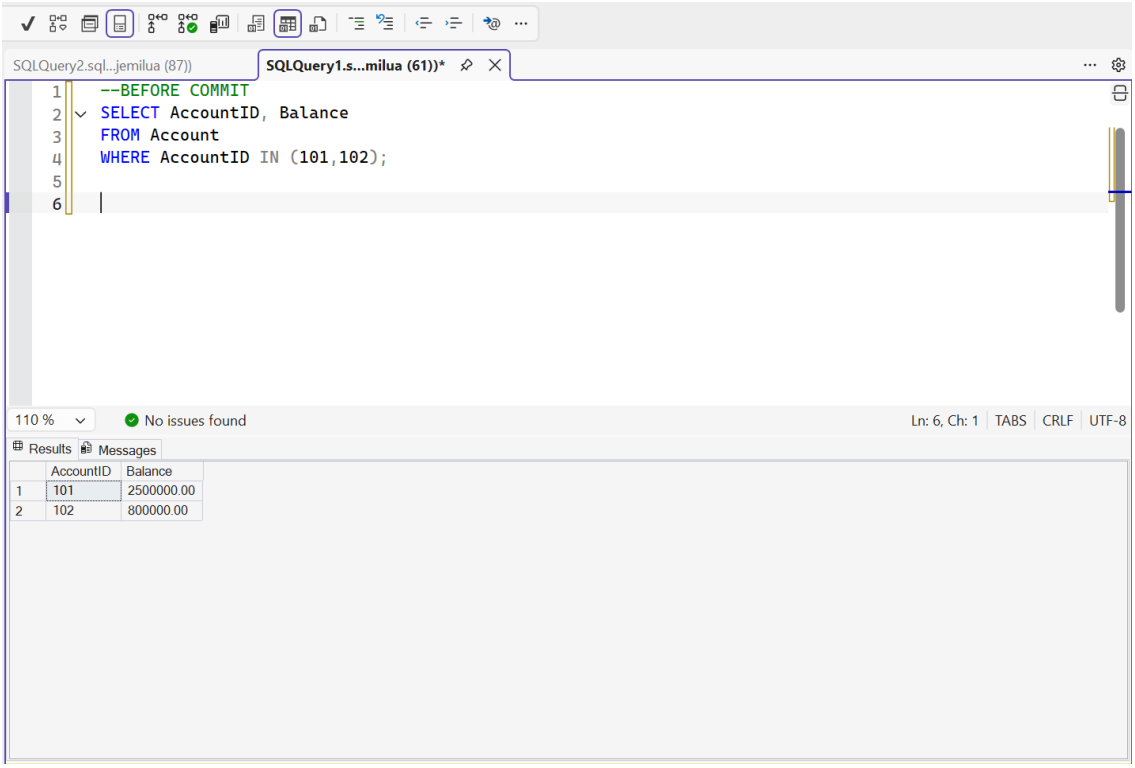
Git Hub Repository url: https://github.com/TosinKyle/MIT8103-Advanced-Database-Systems-Apex-Digital-Bank

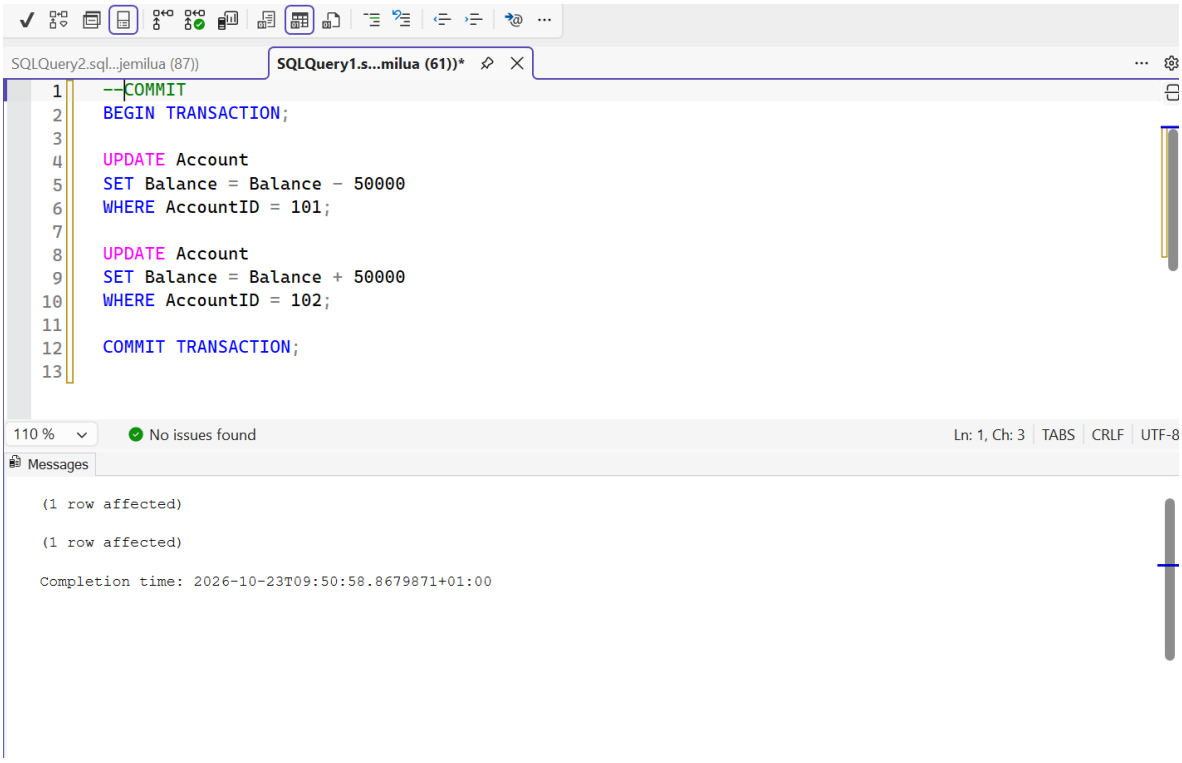
PORTFOLIO 3: TRANSACTIONS AND CONCURRENCY

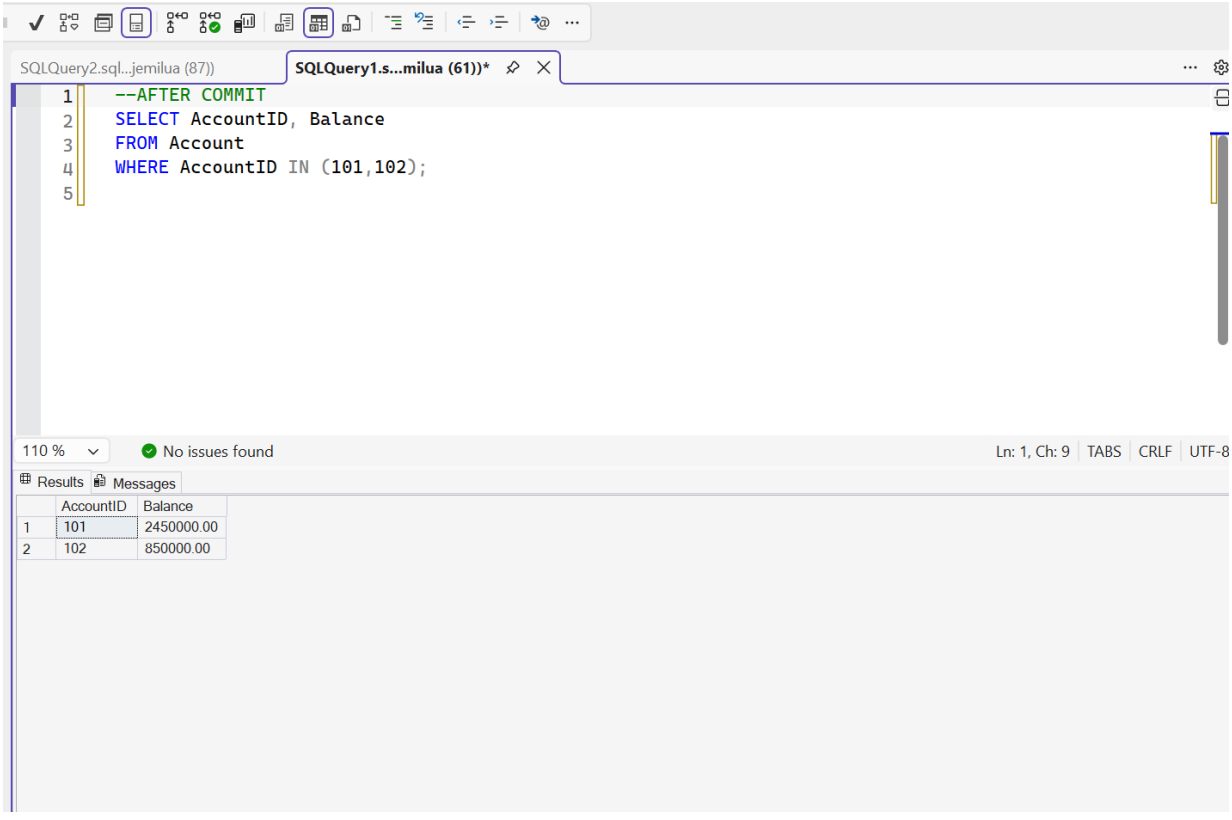
❖ OBJECTIVE

To demonstrate the implementation of transaction management and concurrency control mechanisms within the Apex Digital Bank database using SQL Server, ensuring data consistency, integrity, and reliability during simultaneous database operations.

❖ COMMIT TRANSACTION EXAMPLE




SQLQuery2.sql...jemilua (87))
SQLQuery1.s...milua (61))*
--COMMIT
BEGIN TRANSACTION;
UPDATE Account
SET Balance = Balance - 50000
WHERE AccountID = 101;
UPDATE Account
SET Balance = Balance + 50000
WHERE AccountID = 102;
COMMIT TRANSACTION;
110 %
No issues found
Ln: 1, Ch: 3
TABS
CRLF
UTF-8
Messages
(1 row affected)
(1 row affected)
Completion time: 2026-10-23T09:50:58.8679871+01:00


SQLQuery2.sql...jemilua (87))
SQLQuery1.s...milua (61))*
--AFTER COMMIT
SELECT AccountID, Balance
FROM Account
WHERE AccountID IN (101,102);
110 %
No issues found
Ln: 1, Ch: 9
TABS
CRLF
UTF-8
Results
Messages
AccountID
Balance
1
101
2450000.00
2
102
850000.00

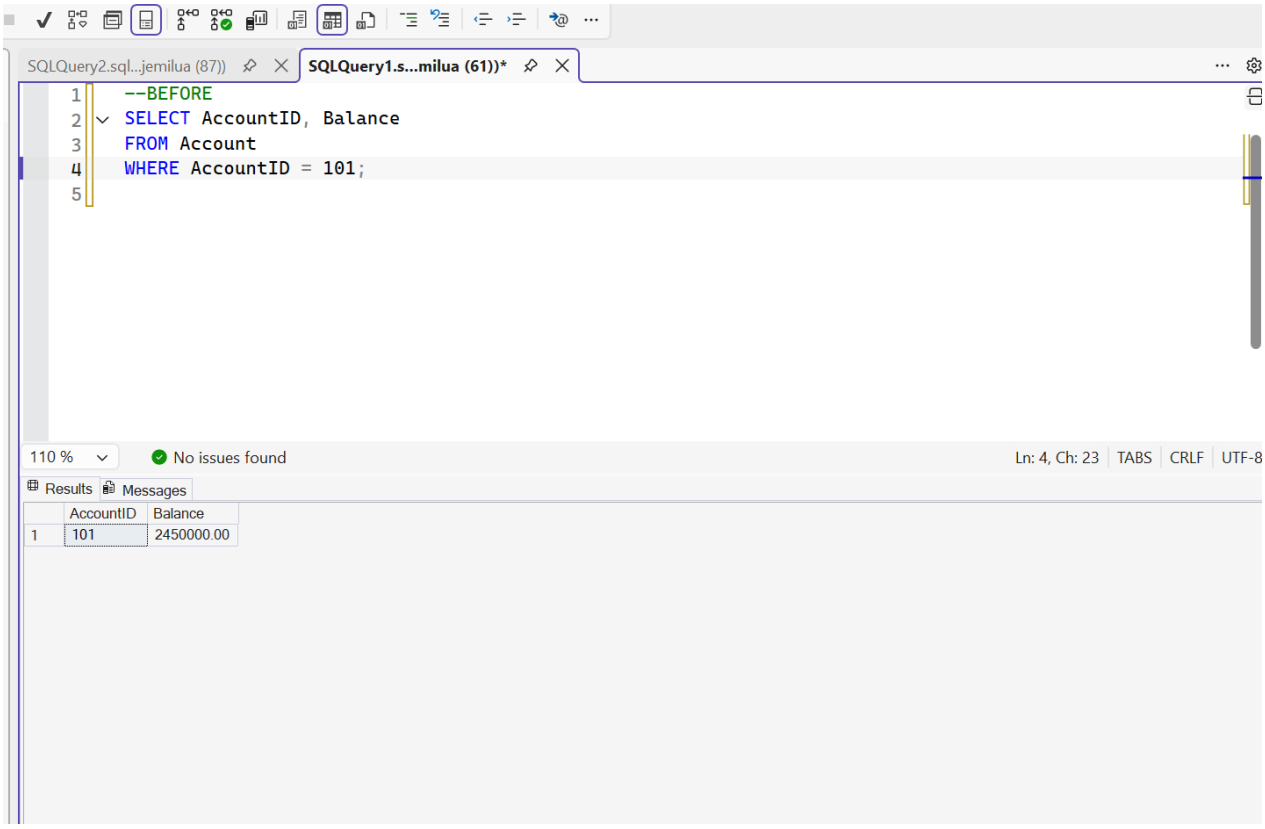
Explanation

From the screenshots above, a transfer of ₦50,000 was performed between two customer accounts. The COMMIT statement permanently saved the updates, demonstrating successful transaction completion. We can see before the commit, the commit and after the commit.

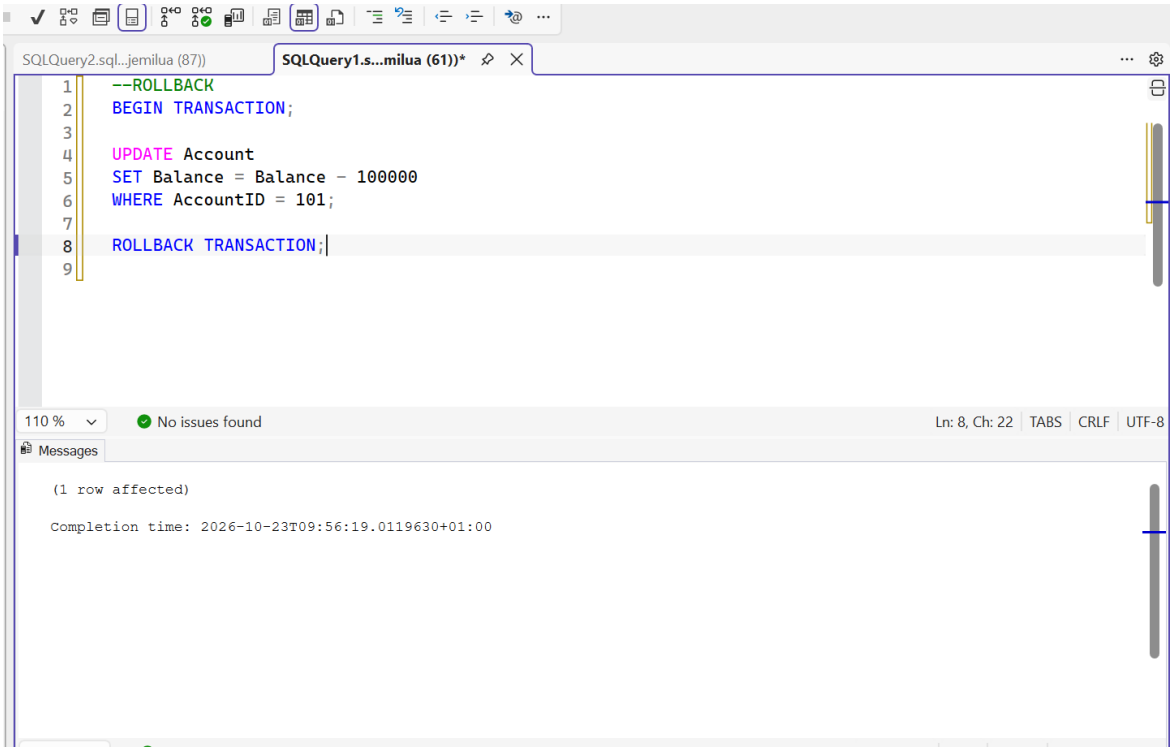
The database state remains consistent.

❖ ROLLBACK TRANSACTION EXAMPLE

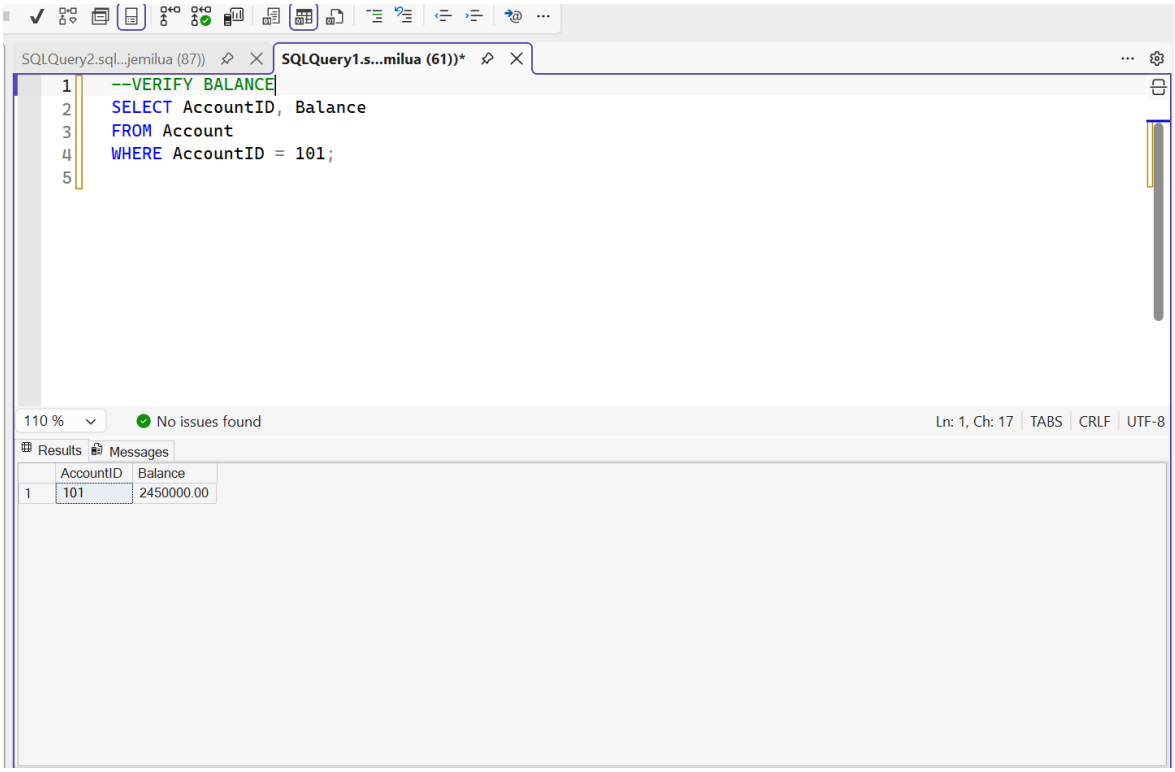
BEFORE CHECK BALANCE



ROLLBACK TRANSACTION



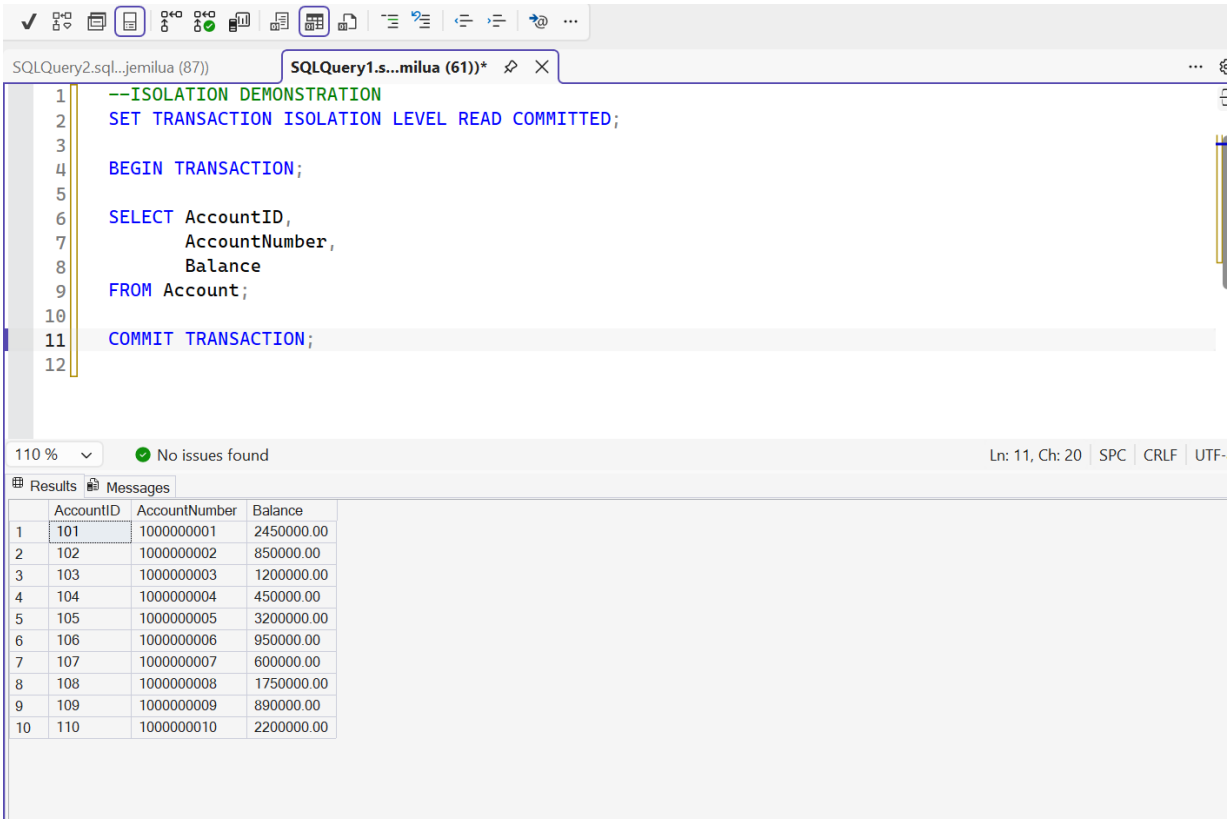
VERIFY BALANCE



Explanation

The transaction was cancelled using ROLLBACK. SQL Server reversed the update and restored the original account balance. Database state remains consistent.

❖ ISOLATION LEVEL DEMONSTRATION

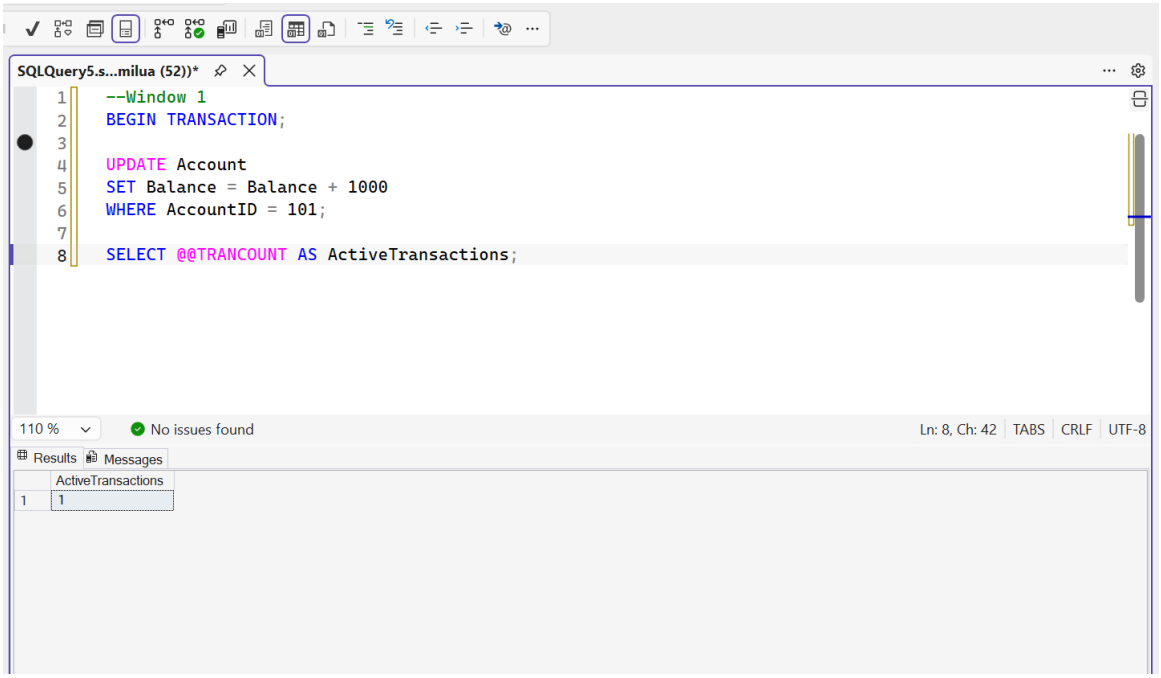


	AccountID	AccountNumber	Balance
1	101	1000000001	2450000.00
2	102	1000000002	850000.00
3	103	1000000003	1200000.00
4	104	1000000004	450000.00
5	105	1000000005	3200000.00
6	106	1000000006	950000.00
7	107	1000000007	600000.00
8	108	1000000008	1750000.00
9	109	1000000009	890000.00
10	110	1000000010	2200000.00

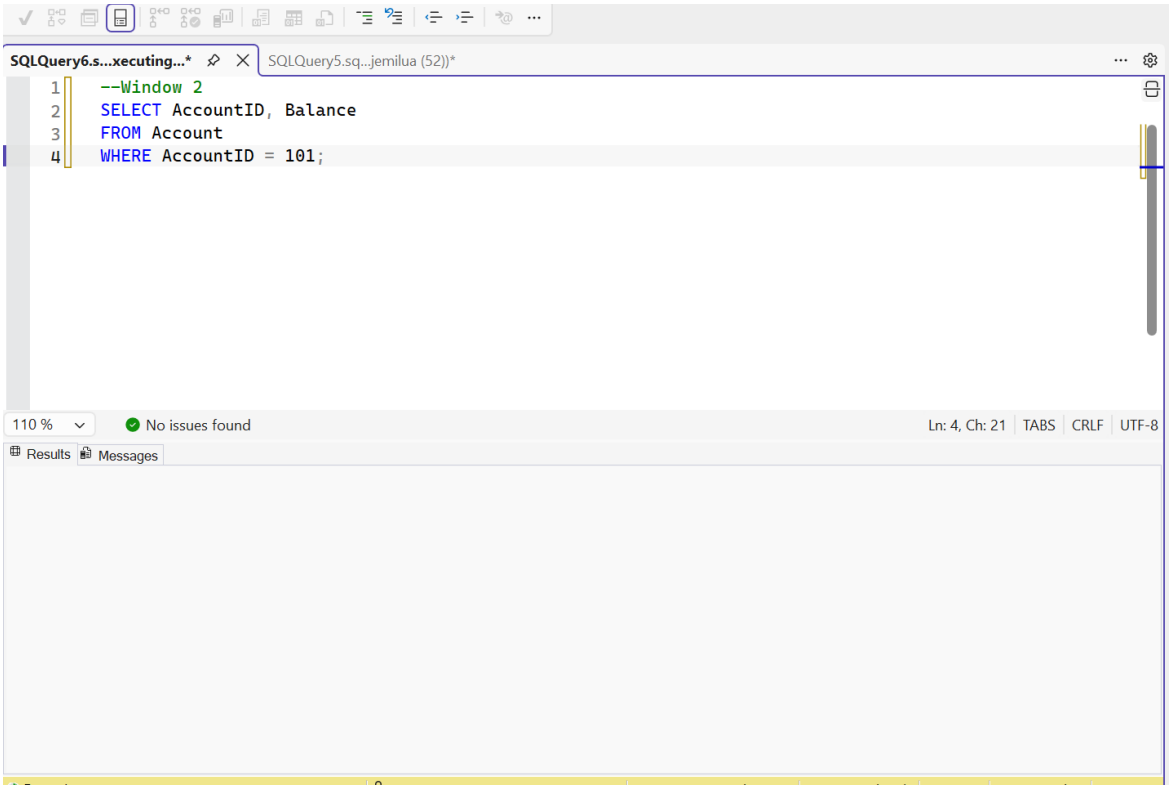
Explanation

The READ COMMITTED isolation level was used to ensure that only committed data could be accessed by transactions, preventing dirty reads.

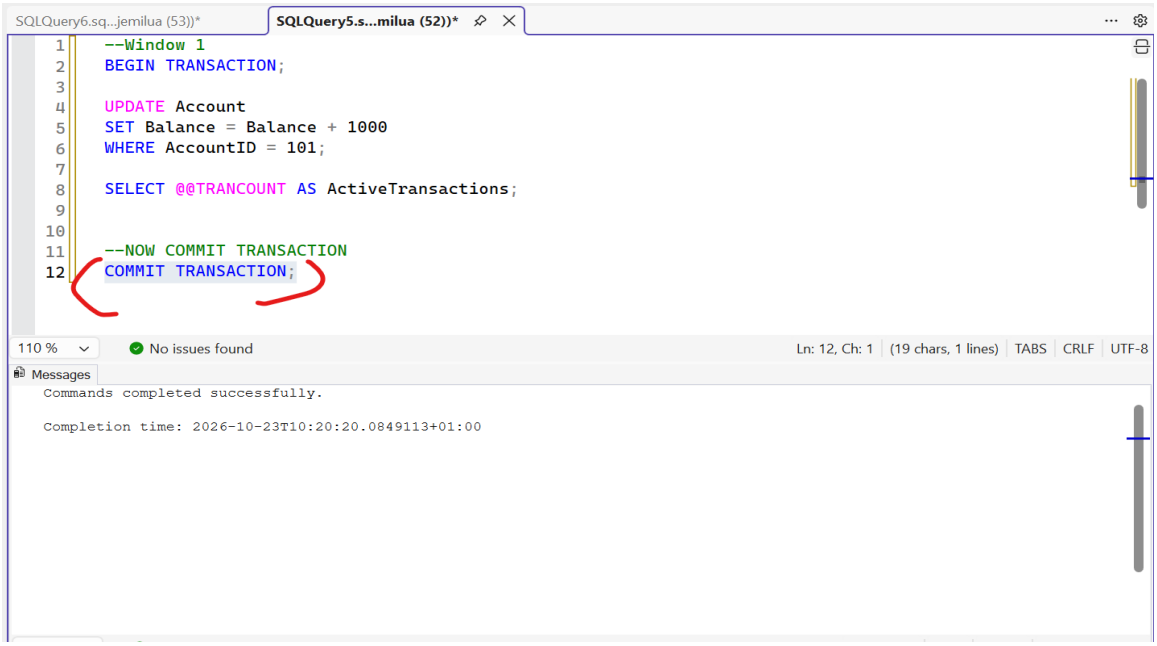
❖ CONCURRENCY CONTROL DEMONSTRATION



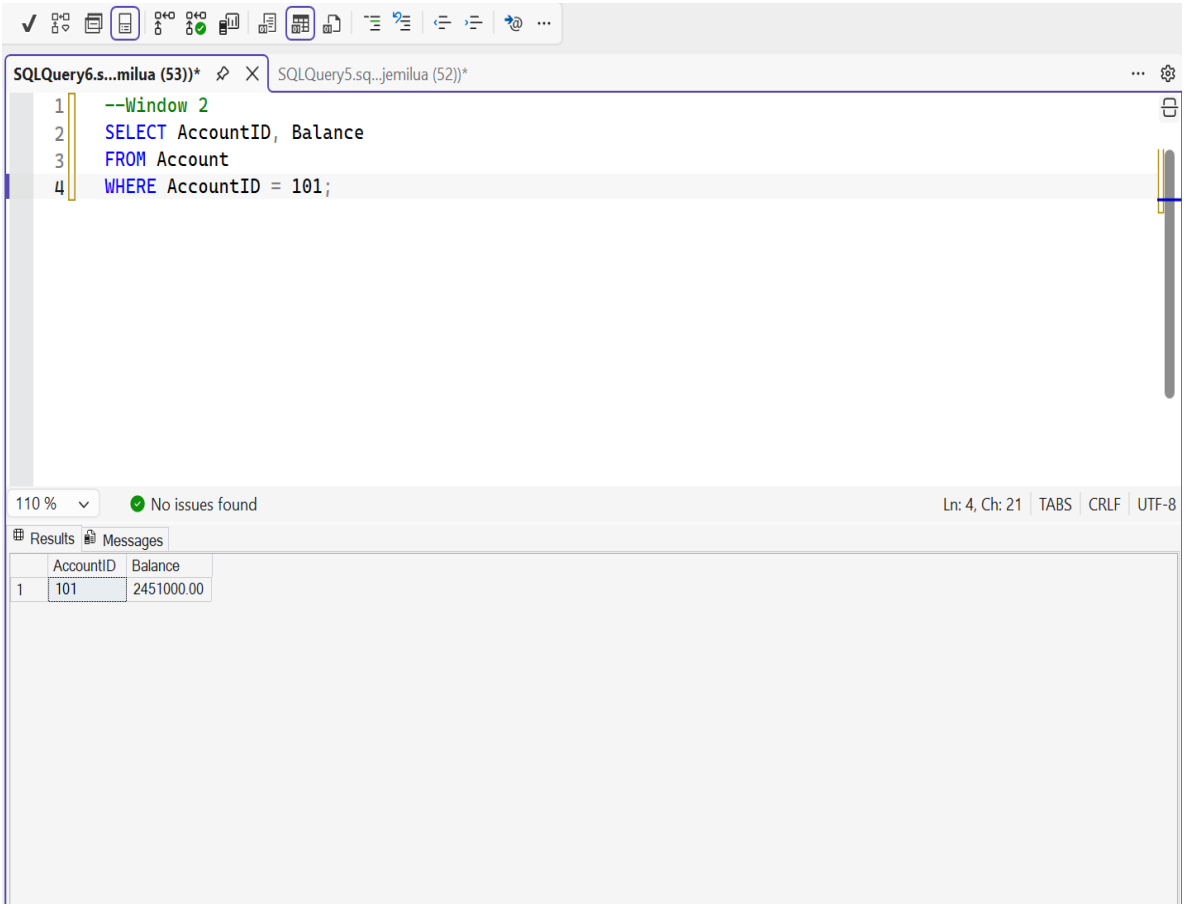
Window 2 – Executing and taking time because the update acquired a lock on AccountID 101.



Transaction committed now



Now the lock is released, other transactions can access the row



Explanation

A concurrency control test was performed using two SQL Server sessions. The first session initiated a transaction and updated an account record without committing the change. When the second session attempted to access the same record, SQL Server delayed execution because the resource was locked by the active transaction. Once the transaction was committed, the lock was released and access was granted. This demonstrates how SQL Server uses locking mechanisms to prevent conflicting operations and maintain data consistency during concurrent access.

❖ ACID PROPERTIES

Atomicity

The transfer transaction was executed as a single unit of work. Either all changes were committed or all changes were rolled back.

Consistency

Database rules and account balances remained valid before and after transaction execution.

Isolation

Concurrent transactions were isolated through SQL Server locking and transaction isolation mechanisms.

Durability

Once committed, transaction updates remained permanently stored within the database.

❖ CONCLUSION

Portfolio 3 has demonstrated the application of transaction management and concurrency control techniques within the Apex Digital Bank database. COMMIT and ROLLBACK operations ensured transactional reliability, while SQL Server locking and isolation mechanisms-maintained data consistency during concurrent access to the database. The implementation of ACID principles contributed to the integrity and reliability of banking operations.